

Bandwidth (MB/s, log)

DRAM tier

NVMe tier

DRAM → NVMe Seq: ~18× slower

**~400×
slower
(rand 4K)**

GPU DRAM
(D2D)

UMA H2D

UMA D2H

CPU DRAM
(memcpy)

NVMe Seq Read
(256K)

NVMe Seq Write
(256K)

NVMe Rand Read
(4K)

NVMe Rand Write
(4K)

22.9 GB/s

25.8 GB/s

25.0 GB/s

6.0 GB/s

1.4 GB/s

1009 MB/s

62 MB/s

129 MB/s